

Supplementary Material

Mamba3Yolo: Mamba-3 Enhanced Real-Time Object Detection with Explainability and Edge Quantization

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1 Additional Implementation Details

1.1 Mamba3ODSSBlock Hyper-parameters

Default settings used in all experiments (Tiny scale):

Hyper-parameter	Value
d_{state}	64
expand ratio	2
headdim	64
MIMO rank r	4
DropPath rate	0.0 – 0.1 (linearly increasing with depth)
LSBlock kernel	3×3 depthwise
RGBlock expansion	2.0

1.2 Training Recipe

- Optimizer: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Initial LR: 1×10^{-3} , cosine decay to 1×10^{-5}
- Weight decay: 0.05
- Epochs: 300 (COCO), 100–150 (medical multi-domain)
- Batch size: 16–64 (depending on GPU memory)
- Augmentations: Mosaic (prob 1.0), MixUp (0.1), random affine, HSV, horizontal flip
- EMA decay: 0.9999
- Loss: CIoU + DFL + BCE (classification) with Task-Aligned Assigner (final version)

1.3 Quantization Protocol

1. Collect activation statistics on 256 images from the validation set.
2. Apply per-token static scaling for Mamba-3 layers (following PTQ4VM).
3. SmoothQuant-style migration of outliers for Linear layers.
4. Export to ONNX (opset 17) then build TensorRT engine with `--fp16` or `--int8 + calibrator`.

2 Additional Ablations

2.1 Effect of Scan Directions

We tested 1, 2, and 4 directional scans inside Mamba3SS2D. Four directions give the best AP_s (+0.7 over two directions) at a modest latency cost (+4 %).

2.2 MIMO Rank Sensitivity

MIMO rank r	mAP	Params	FPS
1 (SISO)	45.3	6.1M	191
2	45.6	6.4M	188
4 (default)	45.9	6.8M	185
8	46.0	7.5M	179

3 Extra Qualitative Results

3.1 Failure Cases

Typical failure modes remaining after Mamba-3 upgrade:

- Extremely small objects ($< 8 \times 8$ px) under heavy occlusion
- Motion blur in endoscopic sequences
- Domain shift to completely unseen imaging devices (different color response)

3.2 Additional Grad-CAM++ Examples

(Insert high-resolution figures generated by `scripts/generate_xai_figures.py` here.)

[Figure S1: Grad-CAM++ on polyp, blood-cell and fundus images]

4 Broader Impact & Limitations

Mamba3Yolo is intended for research and clinical decision support, not as a standalone diagnostic device. All medical results should be validated under local regulatory frameworks before deployment. The model may inherit biases present in the public training sets (e.g., under-representation of certain skin tones or equipment vendors). We encourage users to fine-tune on local data and to keep a human expert in the loop.

5 Reproducibility Checklist

- Code: <https://github.com/ShMazumder/Mamba3Yolo>
- Configs: `configs/models/`, `configs/datasets/`
- Exact package versions: `requirements.txt`
- Random seed: 42 (all experiments)
- Hardware for reported numbers: NVIDIA T4 (Kaggle), RTX 3090, Jetson Orin NX

6 Societal Impact Statement

Improved real-time, explainable detectors can lower the barrier for AI-assisted screening in low-resource clinics. At the same time, over-reliance on automated systems without proper validation can lead to missed diagnoses. We release the full pipeline (including quantization and XAI) so that the community can audit, improve, and adapt the method responsibly.